SOP - TechnicalDiagnosisAndAuditLogging - Parent

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Document ID:     SOP - TechnicalDiagnosisAndAuditLogging - Parent
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        SOP (Standard Operating Procedure)
Purpose:         TechnicalDiagnosisAndAuditLogging
Tier / Level:    Parent (Master Diagnostic Protocol)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Lead TechOps Specialists, Systems Support Leads, E-Commerce Operations Directors
Ecosystem Scope: Shopify Admin (Order Timeline & Event Log), Chrome DevTools Network Tab, Shopify Payments / Stripe
Gateway, Shopify Discounts Engine
Business Unit:   Synapse Botanicals (Technical Operations & Storefront Diagnostics)
```

1. Purpose, Scope & The "3-Minute Diagnostic" Framework

1.1 Purpose & Triage Philosophy

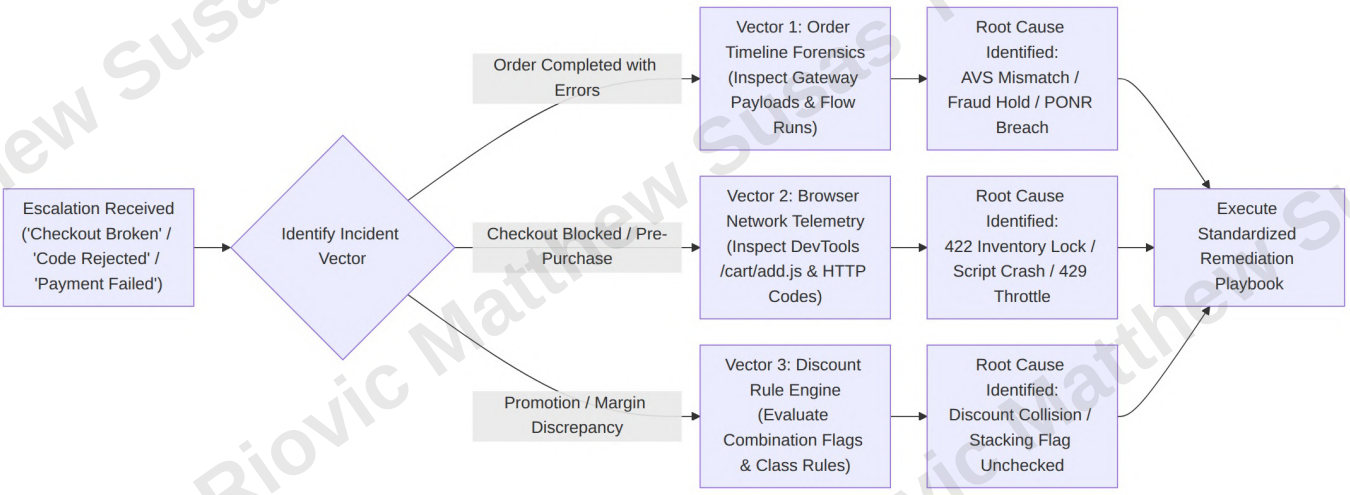
In high-velocity direct-to-consumer (DTC) e-commerce, technical incidents manifest rapidly: a customer cannot add a product to their cart, a payment gateway authorization fails cryptically, an influencer discount code throws a checkout error, or a marketing automation inadvertently destroys unit economics by stacking overlapping promotions.

When these escalations occur, junior operators often guess, clear browser cookies needlessly, or escalate tickets to engineering without evidence. A senior TechOps specialist operates as a **Shopify Detective**, leveraging deterministic diagnostic tools to isolate the exact root cause in **under 3 minutes**.

This Standard Operating Procedure (SOP) defines the mandatory forensic protocols across three diagnostic vectors:

1. **The Order Timeline as Immutable Single Source of Truth:** Extracting raw payment gateway transaction payloads, AVS/CVV risk codes, and automated mutation receipts.
2. **Storefront Browser DevTools & Network Telemetry:** Capturing live AJAX Cart API failures (`/cart/add.js`, `/cart/change.js`), evaluating HTTP response status codes (`422 Unprocessable Entity`, `429 Too Many Requests`), and identifying crashing third-party scripts.
3. **Discount Stacking & Combination Governance:** Resolving discount collision conflicts, managing Shopify's three discount classes (Product, Order, Shipping), and enforcing margin protection floors.

1.2 The 3-Minute Triage Architecture



2. The Order Timeline Forensic Standard

The Shopify Order Timeline is an append-only, tamper-evident chronological ledger. It records every native action, payment transaction, fraud assessment, app mutation, and customer notification associated with an order.

2.1 Anatomy of an Order Timeline Event

Every event on the timeline contains four forensic metadata attributes:

1. **Precise UTC Timestamp:** The exact millisecond the transaction or mutation resolved.
2. **Actor Classification:** Distinguishes between:
   - **System:** Native Shopify core engine actions (e.g., inventory reservations, payment settlement).
   - **Customer:** Storefront actions initiated via checkout sessions.
   - **App / API Client:** Programmatic actions performed via Admin API (e.g., Gorgias CRM, ShipBob WMS, Klaviyo).
   - **Staff User:** Manual actions executed by logged-in human operators (includes staff email address).
3. **Transaction Details Modal:** Collapsible technical metadata containing raw gateway strings and API response objects.
4. **Visibility Scope:** Public customer communication (emails/SMS sent) vs. internal operational notes.

2.2 Inspecting Gateway Payment Transactions & Raw Gateway Strings

When a payment fails, succeeds partially, or triggers an unexpected authorization hold, operators must not rely on the high-level summary. They must inspect the underlying gateway authorization payload:

```
Step-by-Step Gateway Inspection Protocol:
1. Open Order in Shopify Admin.
2. Scroll to the "Timeline" section at the bottom of the page.
3. Locate the payment entry: "A $XX.XX USD payment was processed on [Gateway]..."
4. Click the small disclosure arrow [ > ] to expand the transaction summary.
5. Click "Gateway Information" or "Transaction details" to view the raw JSON payload.
```

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Raw Gateway Transaction Payload (Shopify Payments / Stripe Backend)


```
{
  "id": "ch_3Nxy82LK41890aB1089201a",
  "object": "charge",
  "amount": 8000,
  "amount_captured": 8000,
  "currency": "usd",
  "payment_method_details": {
    "card": {
      "brand": "visa",
      "last4": "4242",
      "country": "US",
      "exp_month": 12,
      "exp_year": 2028,
      "checks": {
        "address_line1_check": "unchecked",
        "address_postal_code_check": "fail",
        "cvc_check": "pass"
      }
    },
    "type": "card"
  },
  "outcome": {
    "network_status": "approved_by_network",
    "reason": null,
    "risk_level": "elevated",
    "risk_score": 68,
    "seller_message": "Shopify Payments: Elevated risk detected due to postal code mismatch.",
    "type": "manual_review"
  },
  "status": "succeeded"
}
```


Forensic Interpretation Guidelines:

- **`network_status` vs. `status`:** Notice above that the issuing bank *approved* the transaction (`approved_by_network`), and the charge *succeeded* technically, but the fraud engine flagged it for `manual_review` because `address_postal_code_check` was `"fail"`.
- **AVS Postal Check:** Indicates that the cardholder entered a billing zip code that does not match the file on record at the card issuer (common indicator of card-not-present fraud or customer relocation).

> **NOTE System Architecture Dependency (Order Lifecycle Holds):**
> When a gateway transaction evaluates to elevated risk with an AVS postal code failure, Shopify Flow automatically applies the `flow-hold:fraud-review` tag. For details on how this hold suspends 3PL fulfillment, refer to **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 2.2**.

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2.3 Identifying App vs. Flow vs. Human Actions in the Timeline

A major source of confusion during operational audits is determining *who* modified an order. The table below provides the visual forensic indicators used to distinguish actors in the Order Timeline:

Actor Type	Visual Indicator in Timeline	Raw Event Metadata	Typical Operational Scenarios
Native System	Shopify bag icon (`Shopify Core`)	`event.app_id: null` `event.author: "Shopify"`	Payment capture, automatic tax calculation, native shipping rate calculation.
Shopify Flow	Flow lightning bolt icon	`event.author: "Shopify Flow"` `event.message: "Flow [Name] ran"`	Automated tagging (`flow-hold:fraud-review`), automatic fulfillment hold injection.
Third-Party App	App logo or generic API plug icon	`event.app_id: 1829410` `event.client_id: "shipbob_wms"`	3PL order ingestion, tracking number synchronization, Gorgias ticket creation.
Human Staff	User avatar / initials with staff name	`event.user_id: 88129410` `event.email: "ops@synapse.com"`	Manual refunds, address overrides, manual line-item swaps, hold releases.

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3. SOP - TechnicalDiagnosisAndAuditLogging - Protocol A: Browser DevTools & Storefront Network Telemetry

When a customer reports: "*I can't add items to my cart*", "*The checkout button is spinning indefinitely*", or "*My screen goes blank when I enter a discount*", the root cause is never visible in Shopify Admin. It lives in the client browser's JavaScript execution runtime and HTTP network stream.

3.1 The Cart API Lifecycle & Endpoint Architecture

The Synapse Botanicals storefront communicates with Shopify's native checkout and cart engines using four standard JSON endpoints:

Endpoint Path	HTTP Method	Payload Parameters	Purpose & Operational Function
`/cart/add.js`	`POST`	`id` (Variant ID), `quantity`, `properties`	Ingests a variant into the current session's shopping cart.
`/cart/change.js`	`POST`	`id` (or `line`), `quantity`	Updates line item quantities (increment, decrement, or remove).
`/cart/clear.js`	`POST`	None	Empties the shopping cart session completely.
`/cart.js`	`GET`	None	Retrieves current cart state, total price, discounts, and item array.

3.2 HTTP Status Code Matrix for Storefront Operations

TechOps specialists must inspect the **Network** tab in Chrome DevTools to triage failed API requests:

```
DevTools Network Inspection Procedure:
1. Open Google Chrome in Incognito mode.
2. Navigate to https://synapsebotanicals.com.
3. Press [F12] or [Ctrl+Shift+I] (Cmd+Option+I on Mac) to open DevTools.
4. Select the "Network" tab.
5. In the filter box, type "cart" and select the "Fetch/XHR" filter pill.
6. Re-create the customer's action (e.g., click "Add to Cart" on a PDP).
7. Inspect the resulting request entry in the table.
```

HTTP Status Code	Response Classification	Root Cause at Synapse Botanicals	Storefront Manifestation	Immediate TechOps Action
`200 OK`	Nominal Success	Request accepted and processed cleanly by Shopify cart engine.	Cart slide-out opens; line item appears with updated subtotal.	None required. Cart state healthy.
`422 Unprocessable Entity`	Business Logic Rejection	**Inventory Ceiling Breached:** Attempting to add an out-of-stock SKU, or requested quantity exceeds available inventory across all nodes.	Button text changes to "*Sold Out*" or displays inline warning toast.	Verify Available-to-Sell (ATS) inventory balance in Shopify Admin for the variant SKU.
`429 Too Many Requests`	Rate Limit / Bot Block	Client IP triggered Shopify's automated bot protection or rate-limiting filter.	Page freezes or returns a Cloudflare / Shopify captcha challenge.	Verify if the user is behind a corporate VPN or executing rapid automated script requests.
`400 Bad Request`	Client Payload Error	Malformed JSON request body or missing mandatory `id` parameter.	Nothing happens on click; console logs uncaught JavaScript error.	Inspect theme JavaScript (`theme.js` / cart drawer component) for broken serialization.
`500 Server Error`	Script / Theme Crash	Third-party app snippet crashed within the theme runtime (e.g., upsell app blocking AJAX chain).	Cart button spins forever; cart drawer never opens.	Inspect **Console** tab for `Uncaught TypeError` originating from third-party app bundles.

Real-World 422 Unprocessable Entity Payload Inspection

When an item is sold out or locked, `/cart/add.js` returns an explicit error payload:

```
{
  "status": 422,
  "message": "You can't add more Synapse Adaptogen Tincture (60ml) - AM Focus to the cart.",
  "description": "All 0 remaining units are currently allocated to pending checkouts or fulfillments.",
  "error": "out_of_stock"
}
```

Diagnostic Verdict: This is not a "broken checkout" or theme bug; it is an inventory depletion event. The customer is attempting to buy units of `SYN-TNCT-AM` when available-to-sell inventory at `LOC-NV-02` is `0`.

3.3 Terminal cURL Diagnostic Probe

To bypass client-side theme issues and directly verify whether a variant ID can be added to a cart via raw Shopify API:

```
# Diagnostic Cart Injection Probe:
curl -s -X POST synapse-botanicals                                    \
  -H "Content-Type: application/json" \
  -H "Accept: application/json" \
  -d '{
    "items": [
      {
        "id": 482194120912,
        "quantity": 1
      }
    ]
  }' | jq .
```

- If this returns `{"items": [...]}` with HTTP `200`, the Shopify backend is functioning properly, proving the bug resides in the storefront JavaScript theme layer.
- If this returns `{"status": 422}`, the issue is confirmed as an inventory or catalog configuration error in the Shopify Admin.

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4. SOP - TechnicalDiagnosisAndAuditLogging - Protocol B: Payment Gateway Authorization & Fraud Diagnostic Protocol

When a customer contacts support claiming their payment was declined, TechOps must rapidly identify whether the failure is a **Soft Decline** (customer or bank retryable) or a **Hard Decline** (terminal failure requiring an alternate card).

4.1 Payment Failure Taxonomy & Response Codes

Shopify Admin Enum & Raw Gateway Code	Failure Class	Detailed Technical Root Cause	Customer Resolution Script	TechOps Remediation
`DO_NOT_HONOR` `(do_not_honor)`	Hard Decline	**The #1 Bank Decline:** Issuer rejected authorization without sharing private financial details.	"*Your card issuer declined the charge with a 'Do Not Honor' notice. Please contact your bank to authorize transactions from Synapse Botanicals, or try an alternate card.*"	Merchant cannot override bank fraud/credit algorithms. Direct buyer to issuing bank.
`AUTHENTICATION_REQUIRED` `(authentication_required)`	Soft Decline	**3D Secure / SCA Failure:** Customer failed or closed the SMS OTP / banking app biometric verification window.	"*Your bank required identity verification (3D Secure) that was not completed. Please try again and complete the security prompt sent to your phone or banking app.*"	Verify that Shopify Payments 3DS settings are active and edge WAF rules do not block 3DS iframes.
`INSUFFICIENT_FUNDS` `(insufficient_funds)`	Soft Decline	Available account balance or credit line cannot cover the charge total.	"*The transaction could not be completed due to insufficient funds. Please check your account balance or select another payment method.*"	Suggest splitting order or using an alternative digital wallet (Shop Pay / Apple Pay).
`INCORRECT_CVC` `(incorrect_cvc / invalid_cvc)`	Soft Decline	The 3- or 4-digit security code entered does not match issuer records.	"*The security code (CVV/CVC) was incorrect. Please double-check the back of your card and re-enter your payment details.*"	Check if browser autofill corrupted the security code field.
`EXPIRED_CARD` `(expired_card)`	Hard Decline	The expiration date entered is in the past.	"*The payment card entered has expired. Please use a valid, current card.*"	Guide customer to enter updated card details.
`CARD_DECLINED` `(generic_decline)`	Hard Decline	Generic card refusal by the network or issuing	"*Your payment was declined by the card processor. Please try an*	Direct customer to try digital wallet or

		bank.	*alternate card or payment method.*"	alternate card.
`CALL_ISSUER` `(call_issuer)`	Hard Decline	The issuing bank flagged the transaction and requires verbal authorization from the cardholder.	"*Your bank has requested that you call them directly to approve this transaction before it can proceed.*"	Inform customer this is a standard bank safety hold.
`PICK_UP_CARD` / **`FRAUD_SUSPECTED`** `(stolen_card / lost_card)`	Terminal / Security	The card has been flagged as lost, stolen, or compromised.	"*This card cannot be processed. Please use an alternate payment method.*"	**Immediate Security Action:** Check customer IP, shipping address, and order velocity for card-testing patterns.
`PROCESSING_ERROR` `(processing_error)`	Soft Decline (Transient)	Temporary network timeout between Shopify Payments and card brand rails.	"*A temporary banking network timeout occurred. Please wait 2 minutes and attempt checkout once more.*"	Check `status.shopify.com` for active gateway disruptions.

> **IMPORTANT Checkout Failure vs. Order Timeline Location:**
> If payment fails during initial checkout, Shopify does not generate an Order. The event is recorded under **Orders → Abandoned checkouts → Timeline**. If an authorization capture, post-purchase upsell, or deferred invoice payment fails, it is recorded directly on the active **Order Timeline**.

4.2 Address Verification System (AVS) & CVV Response Code Matrix

Shopify Payments parses AVS and CVV signals from global card networks and presents them across the Admin Fraud Analysis sidebar, the GraphQL `CardPaymentDetails` API, and the Timeline raw transaction JSON.

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AVS (Address Verification System) Matrix

Code (`avsResultCode`)	Timeline JSON (`checks`)	Admin Fraud Analysis Display	Technical Meaning	Risk Assessment & Handling
`Y`	`line1: pass` `postal: pass`	Street and postal code match	Street address and 5-digit ZIP both match.	**Low Risk:** Green-light fulfillment.
`X`	`line1: pass` `postal: pass`	Street and 9-digit ZIP match	Street address and full 9-digit ZIP match.	**Low Risk (Highest Match):** Green-light fulfillment.
`A`	`line1: pass` `postal: fail`	Street address matches; postal mismatch	Street matches; ZIP code fails.	**Medium Risk:** Common when buyer relocates. Permit if CVV is `pass`.
`Z`	`line1: fail` `postal: pass`	Postal code matches; street mismatch	5-digit ZIP matches; street address fails.	**Medium Risk:** Common with PO Boxes or missing Apt/Suite numbers.
`W`	`line1: fail` `postal: pass`	9-digit postal matches; street mismatch	9-digit ZIP matches; street address fails.	**Medium Risk:** Verify apartment/suite number before shipping.
`N`	`line1: fail` `postal: fail`	Neither street nor postal code match	Both street and postal code mismatch bank file.	**High Risk: Flag for Fraud Review (`flow-hold:fraud-review`).** High probability of stolen card.
`G` / `S`	`line1: unavailable` `postal: unavailable`	AVS not supported by issuer	Global/International issuer does not participate in AVS.	**Neutral / Low:** Standard for non-US international orders. Rely on CVV + 3DS.
`D` / `P`	`line1: pass` (D) `postal: pass` (P)	International address / postal match	Street and postal match (D) or postal only (P) for international card.	**Low / Medium Risk:** Standard cross-border verification.
`U` / `R`	`line1: unavailable` `postal: unavailable`	Address information unavailable	System timeout or bank-side verification offline.	**Low Risk:** Transient bank-side unavailability.

CVV (Card Verification Value) Matrix

Code (`cvvResultCode`)	Timeline JSON (`cvc_check`)	Admin Fraud Analysis Display	Technical Meaning	Risk Assessment & Handling
`M`	`"pass"`	CVV is correct	Security code matches bank file.	**Low Risk:** Physical card or authenticated credential verified.
`N`	`"fail"`	CVV isn't correct	Security code entered does not match issuer file.	**CRITICAL RISK: Immediate Hold.** Block fulfillment; customer likely does not have card in hand.
`P`	`"unchecked"`	CVV wasn't available	**Tokenized / Digital Wallet:** Not processed because payment used Apple Pay, Google Pay, or Shop Pay.	**Low Risk:** Normal for biometric/tokenized checkouts. Do NOT flag as missing security code.
`U`	`"unavailable"`	CVV not supported	Issuer is not certified or does not support CVV verification.	**Neutral:** Common with niche corporate cards or international cards.
`S`	`"fail"`	CVV suspicious	Card requires CVV, but merchant request omitted it.	**Medium Risk:** Investigate checkout integration for bypassed input field.

> **TIP Shopify Payments Automated Protection Settings:**
> In Shopify Admin under **Settings → Payments → Shopify Payments (Manage) → Fraud prevention**, merchants can toggle:
>
> 1. "*Decline charges that fail AVS postal code verification*" (Auto-rejects `checks.address_postal_code_check == "fail"`).
> 2. "*Decline charges that fail CVV verification*" (Auto-rejects `checks.cvc_check == "fail"`).
>
> *Caution:* Enabling AVS postal decline will block legitimate customers who recently moved and have not updated their bank billing address.

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5. SOP - TechnicalDiagnosisAndAuditLogging - Protocol C: Discount Stacking, Combinations & Margin Collision Governance

The most financially dangerous technical trap in Shopify occurs when multiple discount promotions collide at checkout. Without rigorous configuration of Shopify's modern **Discount Combinations Engine**, customers can stack multiple codes, driving order contribution margins below zero.

5.1 The Three Modern Discount Classes in Shopify

Shopify categorizes all promotional discounts into three distinct classes:

```
┌──────────────────────────────────────────────────────────────┐
│                    SHOPIFY DISCOUNT CLASSES                  │
├──────────────────────────────┬───────────────────────────────┤
│ 1. PRODUCT DISCOUNTS         │ Applies to specific variants or │
│    (e.g., 20% Off Stick Packs) │ collections. Computed at line level. │
├──────────────────────────────┼───────────────────────────────┤
│ 2. ORDER DISCOUNTS           │ Applies to the aggregate cart subtotal │
│    (e.g., WELCOME15 - 15% Off) │ after product discounts are deducted. │
├──────────────────────────────┼───────────────────────────────┤
│ 3. SHIPPING DISCOUNTS        │ Applies strictly to the shipping rate │
│    (e.g., FREESHIP - Free Grd) │ calculated at checkout.       │
└──────────────────────────────┴───────────────────────────────┘
```

5.2 The Combination Rules Engine (The Combinations Checkbox)

In Shopify, discounts do *not* combine by default. Each individual discount rule contains an explicit **"Combines with"** governance configuration. Operators must configure these settings according to the matrix below:

Discount Class Being Created	Can Combine with: Other Product Discounts?	Can Combine with: Order Discounts?	Can Combine with: Shipping Discounts?	Synapse Botanicals Governance Standard
Product Discounts	**NO** (Unchecked)	**CONDITIONAL**	**YES** (Checked)	Never allow product discounts to stack on other product discounts (prevents double-dipping on sale items).
Order Discounts	**CONDITIONAL**	**NO** (Unchecked)	**YES** (Checked)	Shopify explicitly prohibits two Order-level discounts from combining on the same cart.
Shipping Discounts	**YES** (Checked)	**YES** (Checked)	**NO** (Unchecked)	Free shipping can combine with either one product discount OR one order discount, provided order value exceeds $75.

5.3 The Double-Dip Margin Collision Trap (Worked Example)

To illustrate why discount combinations represent a critical TechOps responsibility, consider the following real-world failure scenario:

```
THE FATAL PROMOTIONAL STACKING COLLAPSE
_____________________________________________________________________

Cart Composition:
  1x The Daily Protocol Starter Bundle (SYN-BNDL-MAO)
  Retail Price: $98.00 | COGS: $26.50 | Shipping Cost: $10.80

Promotion Alpha (Automated):  "Spring Flash Sale" -> 20% Off Sitewide (Product Discount)
Promotion Bravo (Promo Code): "WELCOME15"         -> 15% Off Cart      (Order Discount)
Promotion Charlie (Threshold): "FREESHIP"          -> Free Shipping     (Shipping Discount)

IF ALL THREE ARE PERMITTED TO COMBINE:
1. Product Price:                    $98.00
2. Spring Flash Sale (-20%):        -$19.60 -> Subtotal: $78.40
3. WELCOME15 Code (-15% on $78.40): -$11.76 -> Subtotal: $66.64
4. FREESHIP Code Applied:           -$10.80 absorbed by merchant
_____________________________________________________________________
TOTAL REVENUE COLLECTED:             $66.64
LESS: Product COGS:                 -$26.50
LESS: Paid Carrier Postage (UPS):   -$10.80
LESS: 3PL Pick/Pack Handling:        -$4.50
LESS: Credit Card Processing (2.9%): -$2.23
_____________________________________________________________________
NET CONTRIBUTION MARGIN:             $22.61 (23.0% Margin — Below Minimum 35% Threshold!)
_____________________________________________________________________
```

The Mandatory Margin Protection Standard:

- **The Rule:** Order discounts (`WELCOME15`) must **never** check the box "*Combines with product discounts*" if the store is running an active sitewide sale.
- **The Safeguard:** During active sitewide promotional events, all order-level discount codes must have their combination settings audited and set to:
  `Combines with: None` (or Shipping Discounts only).

---

6. Master Interconnectivity & System Architecture Lineage

This diagnostic SOP connects directly into the surrounding operational matrices and run sheets of the Synapse Botanicals TechOps suite:

Upstream / Downstream Deliverable	Governing Interface / Node	Interconnectivity Mechanism
SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)	Exception Holds & Address Swaps	Timeline diagnostics identify whether an order is in pre-PONR hold or has locked in warehouse fulfillment.
Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)	Physical Warehouse Fulfillment Gates	Timeline transaction entries display the exact moment an order transitioned to `PROCESSING` at `LOC-PA-01` or `LOC-NV-02`.
Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)	Carrier Zone Economics & Margins	Supplies the carrier rate baseline ($10.80 Transcon) utilized in Section 5.3 margin erosion calculations.
Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)	Catalog Schema & Variant Mapping	Verifies that variant IDs targeted in Cart API diagnostic probes match the canonical production catalog.
SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)	Flow Triggers & Fraud Holds	Timeline entries record the firing of Blueprint 1 (Fraud hold) and Blueprint 2 (Heavy-basket tagging) automations.
Matrix - AutomationLogicAndWebhookTriage - Engine (Domain 4.2)	HTTP Status Code Triage	Supplies the standard status code triage definitions (`422`, `429`, `500`) applied to storefront network inspection.
Matrix - DiscountCombinationsAndMarginProtection - Engine (Domain 5.2)	Mathematical Discount Stacking Matrix	Codifies the complete multi-tier permutation matrix and margin protection formulas evaluated in Protocol C.
CaseStudy - DiscountCollisionAndCartDiagnostics - Analysis (Domain 5.3)	Empirical Simulation & Forensic Case Study	Executes live checkout tests of `WELCOME15` vs `FREESHIP`, documenting Order Timeline and Network tab proofs.